

Entropy of pebble and space complexity

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Abstract

Let $\mathcal{L}$ denote the class Logspace and $\mathcal{NL}$ the class NLogspace. We use $\log \mathcal{CFL}$ to denote the closure under logspace reductions of the set of context-free languages. We prove that $\mathcal{NL} \neq \log \mathcal{CFL}$. This result implies $\mathcal{L} \neq \mathcal{P}$ and the stronger separation $\mathcal{NL} \neq \mathcal{P}$.

Keywords: Computational complexity, Logspace, Context-free languages, Entropy, Pebble automata.

This work was created entirely by the sole author; any remaining errors are the sole responsibility of the author. This work received no funding.

1 Introduction

Definition 1. We use $\mathcal{CFL}$ to denote the class of context-free languages. We use $\log \mathcal{CFL}$ to denote the closure of $\mathcal{CFL}$ under logspace reductions.

Let $\mathcal{P}$ denote the class Ptime. We have:

Theorem 2. $\mathcal{NL} \subseteq \log \mathcal{CFL} \subseteq \mathcal{P}$.

Proof. There exist context-free languages that are $\mathcal{NL}$ -complete; see [5]. Then, $\mathcal{NL} \subseteq \log \mathcal{CFL}$.

The class $\mathcal{CFL}$ is contained in $\text{Dtime}(n^3)$; see [6]. Then, $\log \mathcal{CFL} \subseteq \mathcal{P}$. $\square$

The class $\mathcal{NL}$ can be characterised as the union of the levels of the nondeterministic pebble hierarchy; see Theorem 8. We exploit this fact to attack the problem $\mathcal{NL}$ vs. $\log \mathcal{CFL}$. We show that $\mathcal{NL}$ is strictly contained in $\log \mathcal{CFL}$. To this end, we construct a sequence of languages contained in $\mathcal{NL}$, denoted by $\{\text{RA}_k\}_{k \geq 0}$, and satisfying the following properties:

1. There exists a language RA_∞ in $\log \mathcal{CFL}$ such that, for all $m \geq 2$, if RA_∞ belongs to the m -th level of the nondeterministic pebble hierarchy, denoted $\mathcal{NREG}_m$, then the entire sequence $\{\text{RA}_k\}_{k \geq 0}$ is contained in $\mathcal{NREG}_m$. We say that RA_∞ is an *upper bound* for $\{\text{RA}_k\}_{k \geq 0}$ with respect to the nondeterministic pebble hierarchy.
2. The sequence $\{\text{RA}_k\}_{k \geq 0}$ is *high* in the nondeterministic pebble hierarchy, meaning that, for every $m \geq 1$, there exists k such that RA_k does not belong to $\mathcal{NREG}_m$.

It is easy to check that RA_∞ cannot belong to $\mathcal{NL}$.

Lemma 3. *Suppose that $\{\text{RA}_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy and that RA_∞ is an upper bound for this sequence. Then the language RA_∞ does not belong to $\mathcal{NL}$.*

Proof. Suppose that $\{\text{RA}_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy and that $\text{RA}_\infty \in \mathcal{NL}$. Then, there exists m such that $\text{RA}_\infty \in \mathcal{NREG}_m$. This implies that the entire sequence $\{\text{RA}_k\}_{k \geq 0}$ is included in $\mathcal{NREG}_m$. We get a contradiction. The lemma follows. $\square$

We construct $L_{G_0} \in \mathcal{CFL}$ and show that L_{G_0} is an upper bound for every sequence of context-free languages; (see Section 3). This reduces our problem to that of constructing a sequence of context-free languages that is high in the nondeterministic pebble hierarchy. We construct such a sequence, denoted $\{\text{RA}_k\}_{k \geq 0}$, and we show that it is high; (see Sections 4 to 10). We employ entropy arguments to prove this. The connection with entropy is based on the following naive idea: languages that force automata accepting them to store high-entropy random variables are languages that require large work tapes. Let us briefly discuss this issue.

Let $\mathcal{A}$ be a nondeterministic pebble automaton accepting $L \subseteq \Sigma^*$ and $k_{\mathcal{A}}$ the number of pebbles used by $\mathcal{A}$. Let $\mathcal{S}$ be an infinite subset of Σ^* . Let $n \geq 1$. We associate with the triple $(\mathcal{A}, \mathcal{S}, n)$ a random variable $X_{\mathcal{A}}(\mathcal{S}, n)$ that records the locations of $\mathcal{A}$'s pebbles while processing inputs from the set $\mathcal{S} \cap \Sigma^n$. For all $s \geq 0$, if the entropy of $X_{\mathcal{A}}(\mathcal{S}, n)$, denoted $H(X_{\mathcal{A}}(\mathcal{S}, n))$, exceeds $s \cdot \log(n)$ for infinitely many values of n , it follows that $k_{\mathcal{A}} \geq s$.

Let $\Sigma = \{0, 1, \$\}$. Let $k \geq 1$, and let $\mathcal{A}_k$ be a nondeterministic pebble automaton accepting the language $\text{RA}_k \subseteq \Sigma_k^*$. We show that there exists an infinite set $\mathcal{S}_k \subseteq \Sigma_k^*$ such that, for all $d > \frac{1}{8}$ and all $\gamma > 0$, the following inequality holds asymptotically:

$$H(X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)) \geq d \cdot (1 - \gamma) \cdot \log(n).$$

This result entails the highness of $\{\text{RA}_k\}_{k \geq 1}$ and the separation $\mathcal{NL} \neq \log \mathcal{CFL}$.

1.1 Notation and Terminology

Let Σ be a finite alphabet. We use Σ^* to denote the set of all finite words over Σ . We use Σ^+ to denote the set $\Sigma^* - \{\xi\}$, where ξ denotes the empty word.

Let $n \geq 2$, we use Σ^n to denote the subset of Σ^* constituted by the words of length n .

Let $w \in \Sigma^*$. We use $|w|$ to denote the length of w . Given $i < j \leq |w|$, we use $w[i]$ to denote the i -th letter of w , and we use $w[i, \dots, j]$ to denote the factor (infix) $w[i]w[i+1] \cdots w[j]$.

Let $L \subseteq \Sigma^*$, we use $\text{co-}L$ to denote the language

$$\Sigma^* - L = \{w \in \Sigma^* : w \notin L\}.$$

Let $n \geq 1$, we use $\log(n)$ to denote the logarithm of n with respect to the basis 2. We use $\lfloor \log(n) \rfloor$ to denote the lower integer part of $\log(n)$. Notice that, for all $n \geq 1$, the following inequalities hold:

$$\frac{n}{2} \leq 2^{\lfloor \log(n) \rfloor} \leq n.$$

Let G be a digraph, we use $V(G)$ and $E(G)$ to denote the set of nodes and the set of edges of G .

2 $\mathcal{L}$, $\mathcal{NL}$, and the pebble hierarchy

Let $k \geq 0$. Deterministic k -pebble automata are deterministic two-way finite state automata provided with k labeled pebbles, which the finite control can use as markers on the input tape. The study of this restricted type of Turing machine goes back to Kreider and Ritchie [3].

Definition 4. Let $k \geq 0$, a deterministic k -pebble automaton is a tuple

$$\mathcal{M} = (Q, q_0, \Sigma, A, \delta),$$

where:

1. Q is a finite set of states.
2. $q_0 \in Q$ is the initial state.

3. $A \subseteq Q$ is the set of accepting states.
4. Σ is the input alphabet.
5. The transition function

$$\delta : Q \times \Sigma \times (\mathcal{P}(\{1, \dots, k\}))^2 \rightarrow Q \times (\mathcal{P}(\{1, \dots, k\}))^2 \times \{-1, 0, 1\}$$

is deterministic.

Let $\mathcal{M}$ be a k -pebble automaton. Automaton $\mathcal{M}$ is a two-way deterministic finite state automaton provided with k labeled pebbles. This automaton has the following capabilities:

- It can place its pebbles on the tape.
- It can sense the pebbles that lie on the current cell.
- It can pick specific pebbles from the set of pebbles that lie on the current cell.

Let (q, a, B, C) be a tuple such that B and C are disjoint subsets of $\{1, \dots, k\}$. Suppose that $\mathcal{M}$ is processing the input w . Suppose that:

- q is the current inner state.
- a is the character being scanned.
- B is the set of pebbles that are placed on the current cell.
- C is the set of available pebbles.

Suppose $\delta(q, a, B, C) = (p, D, K, \varepsilon)$. Then:

- $B \cup C = D \cup K$,
- $\mathcal{M}$ change its inner state from q to p ,
- $\mathcal{M}$ places the pebbles belonging to D (and only these pebbles) on the current cell, and
- $\mathcal{M}$ moves its head in the direction indicated by $\varepsilon \in \{-1, 0, 1\}$.

Let $\mathcal{M}$ be a deterministic pebble automaton and let w be an input of $\mathcal{M}$. We say that $\mathcal{M}$ accepts w if this input leads the automaton to an accepting state.

Definition 5. We use $\mathcal{REG}_m$ to denote the class of languages accepted by deterministic m -pebble automata.

The *deterministic pebble hierarchy* is the chain of inclusions:

$$\mathcal{REG}_0 \subseteq \mathcal{REG}_1 \subseteq \mathcal{REG}_2 \subseteq \dots$$

We have the following result:

Theorem 6. *The equality*

$$\mathcal{L} = \bigcup_{0 \leq m} \mathcal{REG}_m$$

holds.

Proof. Let $\mathcal{M}$ be an m -pebble automaton. Observe that a single pebble of $\mathcal{M}$ can be simulated using a binary work tape of logarithmic size. Hence, it is straightforward to construct a deterministic Turing machine $\mathcal{N}$ equipped with $m + 1$ work tapes of logarithmic size that simulates $\mathcal{M}$. Therefore,

$$\bigcup_{0 \leq m} \mathcal{REG}_m \subseteq \mathcal{L}.$$

We now prove the converse inclusion. Let $L \in \mathcal{L}$. Then there exist an integer $m \geq 0$ and a deterministic Turing machine $\mathcal{M}$ such that:

- $\mathcal{M}$ accepts L .
- $\mathcal{M}$ is provided with a read-only input tape and m binary tapes of size $\lfloor \log(n) \rfloor - 2$.

Let us prove that each one of those binary tapes can be simulated using three pebbles. Let $w \in \Sigma^n$ be the input of $\mathcal{M}$. Let us focus on the i -th tape and let us consider the configuration reached by this tape at instant t . We represent this configuration as a pair $(1n_L, n_R1) \in (\{0, 1\}^*)^2$. This pair tells us that $n_L n_R$ is the work tape content, and it also tells us that the head assigned to this tape is located on cell $|n_L|$. Let $(n_R1)^R$ be the reverse of n_R1 . Note that $1n_L$ and $(n_R1)^R$ are binary strings whose lengths are bounded above by $\lfloor \log(n) \rfloor - 1$. These binary strings encode two positive integers m_L and m_R that belong to the interval $\{0, \dots, n - 1\}$. We use this to represent the configuration $(1n_L, n_R1)$ using two pebbles that we call 1 and 2. We assign to pebble 1 the position m_L and to pebble 2 the position

m_R . We use these pebbles to simulate the changes occurring on tape i . Let us suppose, for instance, that the transition function of $\mathcal{M}$ forces the head (of tape i) to move one cell to the left, after replacing with 0 the character 1 that was written on the current cell (which is cell $|n_L|$). Let $(1n_L^*, n_R^*1)$ be the configuration reached at time $t+1$. Let m_L^* and m_R^* be the corresponding integers. Notice that

$$m_L^* = \frac{m_L - 1}{2} \text{ and } m_R^* = 2m_R.$$

This means that we have to move pebbles 1 and 2, from the positions m_L and m_R to the positions $\frac{m_L-1}{2}$ and $2m_R$. This can be done with the help of a third pebble. This means that we can simulate this transition using three pebbles. Notice that we can simulate any other transition of the i -th tape using the same three pebbles. This means that we can replace tape i with three pebbles. This also means that we can replace all the m work tapes of $\mathcal{M}$ with $3m$ pebbles. If we do the latter, we obtain a deterministic $3m$ -pebble automaton that accepts L . We get that $L \in \mathcal{REG}_{3m}$. We conclude that $\mathcal{L}$ is included in

$$\bigcup_{0 \leq m} \mathcal{REG}_m.$$

The theorem is proved. $\square$

Let $m \geq 0$. Nondeterministic m -pebble automata are the nondeterministic counterparts of m -pebble automata.

Definition 7. Let $m \geq 0$, we use $\mathcal{NREG}_m$ to denote the class of languages accepted by nondeterministic m -pebble automata.

The *nondeterministic pebble hierarchy* is the chain of inclusions:

$$\mathcal{NREG}_0 \subseteq \mathcal{NREG}_1 \subseteq \mathcal{NREG}_2 \subseteq \dots$$

The proof of the following theorem is completely analogous to the proof of the previous one.

Theorem 8. *The equality*

$$\mathcal{NL} = \bigcup_{0 \leq m} \mathcal{NREG}_m$$

holds.

3 Word morphism and Greibach hardest context-free language

We want to construct a sequence $\{RA_k\}_{k \geq 0} \subseteq \mathcal{CFL}$ and a language $RA_\infty \in \mathcal{CFL}$ such that:

1. $\{RA_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy.
2. RA_∞ is an upper bound for $\{RA_k\}_{k \geq 0}$.

We show that there exists $L_{G_0} \in \mathcal{CFL}$ that is an upper bound for every sequence $\{L_k\}_{k \geq 0} \subseteq \mathcal{CFL}$. This reduces our problem to that of constructing a *high sequence* of context-free languages.

Definition 9. Let Σ and Ω be two finite alphabets. Let f be a function from Σ to Ω^* . Let $\hat{f} : \Sigma^* \rightarrow \Omega^*$ be the function that is defined by the equation

$$\hat{f}(w) = \begin{cases} \varepsilon, & \text{if } w = \varepsilon \\ f(w[1]) \cdots f(w[|w|]), & \text{if } |w| > 0 \end{cases}$$

We say that $\hat{f}$ is a word morphism from Σ^* to Ω^* , and any word morphism between these two free monoids is defined this way from a function $f : \Sigma \rightarrow \Omega^*$.

Definition 10. Let $L \subset \Sigma^*$ and $T \subset \Gamma^*$ be two languages. We say that L is an inverse image of T if and only if there exists a word morphism $\hat{f} : \Sigma^* \rightarrow \Omega^*$ such that $L = \hat{f}^{-1}(T)$.

The class $\mathcal{CFL}$ contains languages that are complete for $\mathcal{CFL}$ under inverse images of *word morphisms*; see [1]. This means that there exists $L_{G_0} \in \mathcal{CFL}$ such that, for all $T \in \mathcal{CFL}$, the language T is an inverse image of L_{G_0} under a word morphism. We use L_{G_0} to denote *Greibach hardest context-free language*; see [1].

We have:

Theorem 11. *Let $k \geq 0$, the set $\mathcal{NREG}_k$ is closed under inverse images of word morphism.*

Proof. Suppose $L \subset \Sigma^*$ is an inverse image of $T \subset \Gamma^*$ and suppose $T \in \mathcal{NREG}_k$. We prove that $L \in \mathcal{NREG}_k$.

Let $\mathcal{M}$ be a nondeterministic k -pebble automaton accepting $T \subset \Gamma^*$. Let $f : \Sigma \rightarrow \Gamma^*$ be a function such that $L = \hat{f}^{-1}(T)$. We construct a

nondeterministic k -pebble automaton $\mathcal{S}$ that receives as input $w \in \Sigma^n$ and simulates the computation of $\mathcal{M}$, on input $\hat{f}(w)$.

Let

$$N_0 = \max \{|f(a)| : a \in \Sigma\}$$

Let Q be the set of states of $\mathcal{M}$. The set of states of $\mathcal{S}$ is equal to

$$Q \times \{0, \dots, N_0\}^{k+1}.$$

We use this set of states to simulate $\mathcal{M}$. Recall that

$$\hat{f}(w) = f(w[1]) \cdots f(w[n])$$

Thus, suppose:

- $\mathcal{M}$ is in state q ,
- the input head is placed on the r -th character of the i -th factor (i.e., factor $f(w[i])$),
- the set of available pebbles is $B \subseteq \{1, \dots, k\}$, and
- given $j \notin B$, the pebble j is placed on the r_j character of the l_j factor.

Then:

- $\mathcal{S}$ is in state $(q, r, s_1, \dots, s_k)$, with:

$$s_i = \begin{cases} r_i, & i \notin B \\ 0, & i \in B \end{cases},$$

- the input head is placed on the i -th cell,
- the set of available pebbles is B , and
- for all $j \notin B$, the pebble j is placed on the l_j -th cell.

The set of accepting states is the set:

$$\{(q, r, s_1, \dots, s_k) : q \in A\}.$$

The transition function is defined accordingly.

It is easy to check that the above construction works. We conclude that $\mathcal{NREG}_m$ is closed under inverse images of word morphism. The theorem follows. $\square$

We obtain the following corollary:

Corollary 12. *Let $m \geq 0$. If $L_{G_0} \in \mathcal{NREG}_m$ the entire class $\mathcal{CFL}$ is included in $\mathcal{NREG}_m$.*

Proof. Suppose $L_{G_0} \in \mathcal{NREG}_m$. Let $L \in \mathcal{CFL}$. There exists a word morphism f such that $L = f^{-1}(L_{G_0})$. The set $\mathcal{NREG}_m$ is closed under inverse images of word morphism. Then, the language L belongs to $\mathcal{NREG}_m$ and the entire class $\mathcal{CFL}$ is included in $\mathcal{NREG}_m$. $\square$

Moreover, we have:

Corollary 13. *There exists a sequence of context-free languages that is high in the nondeterministic pebble hierarchy if and only if the separation $\mathcal{NL} \subset \log \mathcal{CFL}$ holds.*

Proof. Suppose $\mathcal{NL} \subset \log \mathcal{CFL}$. Let $\{L_i\}_{i \geq 0}$ be an enumeration of $\mathcal{CFL}$. This sequence is high in the nondeterministic pebble hierarchy.

The language L_{G_0} is an upper bound for every sequence of languages contained in $\mathcal{CFL}$. Then, if there exists a sequence $\{L_i\}_{i \geq 0} \subseteq \mathcal{CFL}$ that is high in the nondeterministic pebble hierarchy the language L_{G_0} belongs to $\log \mathcal{CFL} - \mathcal{NL}$. $\square$

4 High sequences of context-free languages

We showed that $\mathcal{NL} \neq \log \mathcal{CFL}$ is equivalent to the existence of *high sequences* in $\mathcal{CFL}$. We try an inductive construction of those sequences.

Definition 14. Let Σ, Γ be two finite alphabets and let $\# \notin \Sigma \cup \Gamma$. A bi-string over $(\Sigma, \Gamma, \#)$ is a string

$$w_1 \# r_1 \# \cdots \# w_n \# r_n$$

such that $w_1, \dots, w_n \in \Sigma^*$; and $r_1, \dots, r_n \in \Gamma^*$. We use $s(w, r)$ as a shorthand for the bi-string

$$w_1 \# r_1 \# \cdots \# w_n \# r_n.$$

We say that $w_1, \dots, w_n$ are the *local factors* of $s(w, r)$ and we say that $r_1, \dots, r_n$ are the local co-factors of this bi-string.

Definition 15. Let $T \subset \Sigma^*$ and let $w \in \Sigma^*$. Define:

$$\varepsilon^T(w) = \begin{cases} 1, & \text{if } w \in T \\ 0, & \text{otherwise} \end{cases}$$

We say that $\varepsilon^T(w)$ is the characteristic bit (with respect to T) of the string w .

Definition 16. Let $r = r_1 \cdots r_n$. Let $I = \{i_1, \dots, i_k\}$ be a subset of $\{1, \dots, n\}$. Suppose $i_1 < \dots < i_k$. We use $\bigodot_{i \in I} r_i$ to denote the substring $r_{i_1} \cdots r_{i_k}$.

Definition 17. Let $L \subseteq \Sigma^*$, $T \subseteq \Gamma^*$, and $\# \notin \Sigma \cup \Gamma$. We define the language $T \otimes_\pi L$ as the set of all bi-strings

$$w_1 \# r_1 \# \cdots \# w_n \# r_n$$

that satisfy the condition

$$\bigodot_{i \in T(w)} r_i \in L,$$

where:

$$T(w) = \{i \leq n : \varepsilon^T(w_i) = 1\}.$$

We say that $T \otimes_\pi L$ is the masked concatenation of L over T .

Theorem 18. Let $L, T \in \mathcal{CFL}$. The language $T \otimes_\pi L$ belongs to $\mathcal{CFL}$.

Proof. Let $\mathcal{A}$ and $\mathcal{B}$ be pushdown automata accepting L and T . Let Θ and Δ be the work alphabets of these automata. We suppose that those alphabets are disjoint. We construct a pushdown automaton $\mathcal{N}$ accepting $T \otimes L$. The work alphabet of $\mathcal{N}$ is $\Theta \cup \Delta$.

Let

$$s(w, r) = w_1 \# r_1 \# \cdots \# w_n \# r_n$$

be an input of $\mathcal{N}$. The automaton works on that input as follows:

Let $i \leq n$. Suppose that $\mathcal{N}$ is starting to process the factor w_i . Automaton $\mathcal{N}$ simulates the computation of $\mathcal{B}$ on w_i . We assume:

- At the start of this simulation, the pushdown stack contains a string $X_i \in \Theta^*$. This string X_i coincides with the content of the pushdown store of $\mathcal{A}$ after a run of this automaton on

$$\bigodot_{j \in T(w, i)} r_j,$$

where

$$T(w, i) = \{j < i : \varepsilon^T(w_j) = 1\}.$$

- During the simulation of $\mathcal{B}$ in w_i , the automaton $\mathcal{N}$ cannot access X_i . When the simulation of $\mathcal{B}$ on w_i ends, the automaton $\mathcal{N}$ knows $\varepsilon^T(w_i)$. Then it proceeds as follows:

1. $\mathcal{N}$ pops from the stack all symbols above X_i belonging to Γ .
2. If $w_i \in T$, the automaton $\mathcal{N}$ continues with the simulation of $\mathcal{A}$ while scanning the co-factor r_i . During this phase, $\mathcal{N}$ is allowed to access X_i (i.e., $\mathcal{N}$ is allowed to access all the content of its pushdown store).
3. If $w_i \notin T$, the automaton $\mathcal{N}$ is not allowed to continue with the simulation of $\mathcal{A}$ while scanning the co-factor r_i ($\mathcal{N}$ undergoes a *resting phase* while scanning r_i).

$\mathcal{N}$ accepts $s(w, r)$ if and only if the simulation ends in an accepting state of $\mathcal{A}$.

It can be verified that $\mathcal{N}$ accepts the language $T \otimes_\pi L$. The theorem follows. $\square$

Definition 19. Let L be a context-free language. We now define a sequence $\{L_k\}_{k \geq 0} \subset \mathcal{CFL}$. We define this sequence by induction on k :

1. $L_0 = 1 \cdot \{0, 1\}^*$.
2. Let Σ_k be the alphabet of L_k , let $\#_{k+1} \notin \Sigma_k$. We define

$$L_{k+1} = L_k \otimes_{\#_{k+1}} L.$$

The next proposition is straightforward.

Proposition 20. $\mathcal{NL} \neq \log \mathcal{CFL}$ if and only if there exists $L \in \mathcal{CFL}$ such that $\{L_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy.

Proof. Let L be a language in $\log \mathcal{CFL} - \mathcal{NL}$. This language is logspace reducible to L_1 . Then $L_1 \in \log \mathcal{CFL} - \mathcal{NL}$ and $\{L_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy. On the other hand, if $\{L_k\}_{k \geq 0}$ is high in the pebble hierarchy the language L_{G_0} belongs to $\log \mathcal{CFL} - \mathcal{NL}$. $\square$

5 Entropy of pebble automata

Definition 21. Let X_n be a random variable distributed over the set $\{1, \dots, n\}$. The Shannon entropy of X_n is defined as

$$H(X_n) = - \sum_{i \leq n} \Pr(X_n = i) \log(\Pr(X_n = i)),$$

using the convention

$$0 \cdot \log(0) = 0.$$

A standard fact is that entropy is maximised by the uniform distribution:

- $H(X_n) \leq \log(n)$, and
- the equality holds when X_n is uniformly distributed.

Definition 22. Let X and Y be random variables. We denote by $H(X, Y)$ the entropy of the joint random variable (X, Y) ; this is called the joint entropy of X and Y .

Definition 23. For random variables X and Y the conditional entropy $H(X | Y)$ is defined by

$$H(X | Y) = H(X, Y) - H(Y).$$

See [4] for further foundational material on entropy and information theory.

Definition 24. Let $\mathcal{A}$ be a nondeterministic m -pebble automaton accepting $L \subseteq \Sigma^*$. Let Q denote the set of internal states and let $w \in \Sigma^n$. The *configurations* reached by $\mathcal{A}$ on input w are $(m + 2)$ -tuples in

$$\{0, 1, \dots, n\}^{m+1} \times Q.$$

Let

$$\mathcal{D}_t = (h, p_1, \dots, p_m, q)$$

be one such tuple, representing the configuration reached by $\mathcal{A}$ at time t . We have:

1. h is the current head position.
2. $p_1, \dots, p_{m-1}$, and p_m denote the locations of the m pebbles (if pebble i is not on the tape we set $p_i = 0$).

3. $q \in Q$ is the internal state at time t .

Definition 25. Let $\mathcal{D}_t$ be as above. We call $\mathcal{D}_t$ the instantaneous configuration of $\mathcal{A}$ at time t . Let

$$\mathcal{I}_t = (p_1, \dots, p_m, q).$$

We refer to $\mathcal{I}_t$ as the coding configuration of $\mathcal{A}$ at time t .

Definition 26. Let $L \subset \Sigma^*$ and let $\mathcal{A}$ be a nondeterministic m -pebble automaton accepting L . Let $\mathcal{H}$ be an infinite subset of Σ^* and $n \geq 1$. We define $X_{\mathcal{A}}(\mathcal{H}, n)$ as a random variable uniformly distributed over the set of coding configurations visited by $\mathcal{A}$ during its computations on the elements of $\mathcal{H} \cap \Sigma^n$.

The following theorem is straightforward.

Theorem 27. *Let $\mathcal{A}$ be a k -pebble automaton accepting a language $L \subseteq \Sigma^*$. The following inequality holds asymptotically:*

$$H(X_{\mathcal{A}}(\Sigma^*, n)) \leq (k+1) \cdot \log(n).$$

Proof. Let $\mathcal{A}$ be a k -pebble automaton accepting $L \subseteq \Sigma^*$. The random variable $X_{\mathcal{A}}(\Sigma^*, n)$ is distributed over a subset of $\{1, \dots, n\}^k \times Q$. Then

$$H(X_{\mathcal{A}}(\Sigma^*, n)) \leq k \cdot \log(n) + \log(|Q|).$$

There exists N such that, for all $n \geq N$, the following inequality holds:

$$k \cdot \log(n) + \log(|Q|) \leq (k+1) \cdot \log(n).$$

The theorem follows. □

6 Information theoretic decomposition

Let $L \subseteq \Sigma^*$ be a language in $\mathcal{CFL}$. We study the sequence $\{L_k\}_{k \geq 0}$. The language L_{k+1} is defined by two different constraints:

- the local constraint encoded by L_k and which interacts with the local factors $w_1, \dots, w_n$; and
- the global constraint encoded by L that interacts with the substring $\odot_{i \in L_k(w)} r_i$

We construct a set $\mathcal{H}_{k+1}^\Sigma$ for which these two constraints become *orthogonal* to each other. Over the constructed sets, the entropy of pebble automata accepting L_{k+1} can be decomposed as the sum of two entropies, namely: the entropy of pebble automata accepting L_k and the *cross-entropy* of pebble automata accepting L_1 , (see below). A key element in the construction of the $\mathcal{H}_k^\Sigma$'s is that their constituent bi-strings are made-up of a small (logarithmic) number of factors.

Remark 28. An important remark about the importance of being constituted by a polylogarithmic number of factors. Let $\{X_n\}_{n \geq 1}$ be a sequence of random variables and let $C, D, r > 0$. We have that, for all $\gamma > 0$, the inequality

$$H(X_n) \geq (1 - \gamma) \cdot C \cdot \log(n)$$

holds asymptotically if and only if, for all $\gamma > 0$, the following inequality holds asymptotically:

$$H(X_n) \geq (1 - \gamma) \cdot C \cdot \log(D \cdot \log^r(n) \cdot n).$$

Definition 29. Let Σ , Γ , and $\#$ be as above. Let $\mathcal{H}$ be an infinite subset of Γ^* . Suppose $\mathcal{H} \cap \Gamma^n \neq \emptyset$. We use $\Sigma[\mathcal{H}](3 \cdot \lfloor \log(n) \rfloor \cdot (n + 1))$ to denote the set of all bi-strings $s(w, r)$, over $(\Sigma, \Gamma, \#)$, that satisfy the following four conditions:

1. $s(w, r)$ is equal to:

$$w_1 \# r_1 \# \cdots \# w_{\lfloor \log(n) \rfloor} \# r_{\lfloor \log(n) \rfloor} \#^s.$$

2. For all $1 \leq i \leq \lfloor \log(n) \rfloor$, the string w_i belongs to $\mathcal{H} \cap \Gamma^n$.
3. For all $0 \leq i \leq \lfloor \log(n) \rfloor$, the string r_i belongs to $\Sigma^{\leq n}$.
4. $s = 3 \cdot \lfloor \log(n) \rfloor \cdot (n + 1) - |w_1 \# r_1 \# \cdots \# w_{\lfloor \log(n) \rfloor} \# r_{\lfloor \log(n) \rfloor}|$.

Definition 30. We use $\Sigma[\mathcal{H}]$ to denote the set

$$\bigcup_{n \geq 4} \Sigma[\mathcal{H}](3 \lfloor \log(n) \rfloor (n + 1)).$$

Definition 31. Let Σ be a finite alphabet (the alphabet of L). Define:

1. $\mathcal{H}_0 = \{0, 1\}^+$.
2. $\mathcal{H}_{k+1}^\Sigma = \Sigma[\mathcal{H}_k^\Sigma]$.

Remark 32. Let $L \subseteq \Sigma^*$. Consider the language $L_1 = L_0 \otimes L$ and the set $\mathcal{H}_1^\Sigma(3 \cdot \lfloor \log(n) \rfloor \cdot (n+1))$.

Let

$$s(w, r) = w_1 \#_1 r_1 \#_1 \cdots \#_1 w_{\lfloor \log(n) \rfloor} \#_1 r_{\lfloor \log(n) \rfloor} \#_1^s$$

be a bi-string in $\mathcal{H}_1^\Sigma(3 \cdot \lfloor \log(n) \rfloor \cdot (n+1))$. This bi-string belongs to L_1 if and only if

$$\bigodot_{i: w_i[1]=1} r_i \in L.$$

So, whether $s(w, r)$ belongs to L_1 or not is completely determined by the co-factors $r_1, \dots, r_{\lfloor \log(n) \rfloor} \in \Sigma^*$; and the short binary string $w_1[1] \cdots w_{\lfloor \log(n) \rfloor}[1]$.

Definition 33. Let $\mathcal{A}_1$ be a nondeterministic pebble automaton. We say that $\mathcal{A}_1$ separates $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$ if $\mathcal{A}_1$ accepts all the strings in $L_1 \cap \mathcal{H}_1^\Sigma$ and rejects all the strings in $co-L_1 \cap \mathcal{H}_1^\Sigma$.

Definition 34. Let $\mathcal{A}_1$ be a nondeterministic pebble automaton separating $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$. We use $Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)$ to denote a random variable that is uniformly distributed over the set of coding configurations that are visited by $\mathcal{A}_1$, when this automaton is processing an input $s(w, r) \in \mathcal{H}_1^\Sigma(3 \cdot \lfloor \log(n) \rfloor \cdot (n+1))$ and it is querying a local bit $w_i[1]$.

We have:

Theorem 35. Let $L \subseteq \Sigma^*$ be a context-free language. Let $k \geq 1$ and let $\mathcal{A}_{k+1}$ be a nondeterministic pebble automaton accepting L_{k+1} . There exist nondeterministic pebble automata $\mathcal{A}_k$ and $\mathcal{A}_1$ such that:

1. $\mathcal{A}_k$ accepts the language L_k .
2. $\mathcal{A}_1$ separates $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$.
3. The following equality holds:

$$\begin{aligned} & H(X_{\mathcal{A}_{k+1}}(\mathcal{H}_{k+1}^\Sigma, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1))) \\ &= H(X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)) + H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)). \end{aligned}$$

Proof. Let $\mathcal{A}_{k+1}$ be a nondeterministic pebble automaton accepting L_{k+1} . Let $n \geq 2$ and suppose that $\mathcal{H}_k^\Sigma(n)$ is non-empty. We analyze the entropy of the random variable

$$X_{\mathcal{A}_{k+1}}(\mathcal{H}_{k+1}^\Sigma, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1)).$$

Let $\mathcal{C}_{k+1}$ be an outcome of this variable. We assume that $\mathcal{A}_{k+1}$ is checking a local factor $w_i \in \mathcal{H}_k^\Sigma(n)$ and computing the characteristic bit $\varepsilon^{L_k}(w_i)$. We use $\mathcal{A}_k$ to denote the nondeterministic pebble automaton that accepts L_k and which is used by $\mathcal{A}_{k+1}$ in order to compute the ε^{L_k} -bits. The configuration $\mathcal{C}_{k+1}$ encodes a *local configuration* $\mathcal{C}_k$: the current configuration accessed by $\mathcal{A}_k$ on the local factor $w_i \in \mathcal{H}_k^\Sigma(n)$. The following equality holds:

$$\begin{aligned} & H(X_{\mathcal{A}_{k+1}}(\mathcal{H}_{k+1}^\Sigma, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1))) \\ = & H(\mathcal{C}_k) + H(X_{\mathcal{A}_{k+1}}(\mathcal{H}_{k+1}^\Sigma, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1)) \mid \mathcal{C}_k). \end{aligned}$$

The random variable $\mathcal{C}_k$ is uniformly distributed over the set of coding configurations visited by $\mathcal{A}_k$ on inputs from $\mathcal{H}_k^\Sigma(n)$. This means that $\mathcal{C}_k$ is equivalent to $X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)$ and the following equality holds:

$$H(\mathcal{C}_k) = H(X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)).$$

We observe that:

- $\mathcal{A}_{k+1}$ uses the nondeterministic local subroutine $\mathcal{A}_k$ to compute the ε^{L_k} -bits, and
- it uses a global subroutine $\mathcal{B}_{k+1}$ for computing the next factor to be processed, processing the co-factors, and deciding whether the bi-string $s(w, r)$ is accepted or rejected.

Let us show that we can use $\mathcal{B}_{k+1}$ for separating $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$. Thus, let

$$s(w, r) = w_1 \#_1 r_1 \#_1 \cdots \#_1 w_{\lfloor \log(n) \rfloor} \#_1 r_{\lfloor \log(n) \rfloor} \#_1^s \in \mathcal{H}_1^\Sigma.$$

For all $i \leq n$, we have: $\varepsilon^{L_0}(w_i) = w_i[1]$. We work on this instance as follows:

1. We simulate, on $s(w, r)$, the computation of $\mathcal{A}_{k+1}$ on the string

$$w_1 \#_{k+1} r_1 \#_{k+1} \cdots \#_{k+1} w_{\lfloor \log(n) \rfloor} \#_{k+1} r_{\lfloor \log(n) \rfloor} \#_{k+1}^s.$$

2. Each time $\mathcal{A}_{k+1}$ calls the local subroutine $\mathcal{A}_k$ we proceed as follows: we interrupt the execution of this subroutine (on the current local factor $\#_1 w_i$), ask for the value of $w_i[1]$, and set

$$\varepsilon^{L_k}(w_i) = w_i[1].$$

We use $\mathcal{B}_1$ to denote this truncated version of $\mathcal{B}_{k+1}$. Note that $\mathcal{B}_1$ is a nondeterministic pebble automaton that separates $L_1 \cap \mathcal{H}_1^\Sigma$ from $co\text{-}L_1 \cap \mathcal{H}_1^\Sigma$. Note also that

$$H(Y_{\mathcal{B}_1}(\mathcal{H}_1^\Sigma, n)) = H(X_{\mathcal{A}_{k+1}}(\mathcal{H}_{k+1}^\Sigma, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1)) \mid \mathcal{C}_k).$$

The theorem follows. $\square$

Definition 36. Let $L \subseteq \Sigma^*$ be a language in $\mathcal{CF}\mathcal{L} \cap \mathcal{L}$. We say that L is hard-to-cross if there exists $d > 0$ such that, for all nondeterministic pebble automata $\mathcal{A}_1$ separating $L_1 \cap \mathcal{H}_1^\Sigma$ from $co\text{-}L_1 \cap \mathcal{H}_1^\Sigma$ and all $\gamma > 0$, the following inequality holds asymptotically:

$$H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)) \geq d \cdot (1 - \gamma) \cdot \log(n).$$

We obtain the following corollary:

Corollary 37. Let $L \in \mathcal{CF}\mathcal{L} \cap \mathcal{NL}$ be a hard-to-cross language, the sequence $\{L_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy.

Proof. We show that, for all $\gamma \geq 0$ and all $\mathcal{A}_k$, the following inequality holds asymptotically:

$$H(X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)) \geq k \cdot d \cdot (1 - \gamma) \cdot \log(n).$$

We use induction on k .

The non-negativity of Shannon entropy implies that the inequality holds for $k = 0$.

We assume that the inequality holds for k and show that it holds for $k + 1$. Thus, let $\mathcal{A}_{k+1}$ be a nondeterministic pebble automaton accepting L_{k+1} . Theorem 35 asserts that there exist $\mathcal{A}_k$ and $\mathcal{A}_1$ such that

$$\begin{aligned} & H(X_{\mathcal{A}_{k+1}}(\mathcal{H}_{k+1}^\Sigma, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1))) \\ &= H(X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)) + H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)). \end{aligned}$$

The inductive hypothesis ensures that, for all $\gamma > 0$, the following inequality holds asymptotically:

$$H(X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)) \geq k \cdot d \cdot (1 - \gamma) \cdot \log(n)$$

On the other hand, since L is hard-to-cross, for all $\gamma > 0$, the following inequality holds asymptotically:

$$H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)) \geq d \cdot (1 - \gamma) \cdot \log(n).$$

Then, for all $\gamma > 0$, the inequality

$$H(X_{\mathcal{A}_{k+1}}(\mathcal{H}_{k+1}^\Sigma, n)) \geq (k+1) \cdot d \cdot (1-\gamma) \cdot \log(n)$$

holds asymptotically. We obtain that, for all $k > \frac{1}{d}$, the language L_k does not belong to $\mathcal{NR}\mathcal{EG}_{\lfloor \frac{k}{d} \rfloor}$. This implies that $\{L_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy. $\square$

7 Context-free languages that are hard-to-cross

Corollary 37 reduces the problem of proving $\mathcal{NL} \neq \log \mathcal{CFL}$ to that of constructing a hard-to-cross context-free language.

Let $L \subseteq \Sigma^*$ be a context-free language. We study the *crossing complexity* of separating $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$.

7.1 Entropy and state complexity of context-free languages

Two-way nondeterministic finite state automata (2NFAs) are the same as nondeterministic pebble automata provided with zero pebbles (i.e., 2NFAs are nondeterministic zero-pebble automata).

Definition 38. Let $L \subseteq \Sigma^*$ be a context-free language and $\# \notin \Sigma$.

1. Let $n \geq 2$, we use $POS_n(L)$ to denote the finite set:

$$\left\{ \begin{array}{l} \varepsilon \# r_1 \# \cdots \# r_{\lfloor \log(n) \rfloor} : \\ \varepsilon \in \{0, 1\}^{\lfloor \log(n) \rfloor}, \quad r_1, \dots, r_{\lfloor \log(n) \rfloor} \in \Sigma^{\leq n}; \\ \text{and } \bigodot_{i: \varepsilon[i]=1} r_i \in L \end{array} \right\}.$$

2. We use $NEG_n(L)$ to denote the finite set:

$$\left\{ \begin{array}{l} \varepsilon \# r_1 \# \cdots \# r_{\lfloor \log(n) \rfloor} : \\ \varepsilon \in \{0, 1\}^{\lfloor \log(n) \rfloor}, \quad r_1, \dots, r_{\lfloor \log(n) \rfloor} \in \Sigma^{\leq n}; \\ \text{and } \bigodot_{i: \varepsilon[i]=1} r_i \notin L \end{array} \right\}.$$

Definition 39. Let $\mathcal{A}$ be a 2NFA that separates $POS_n(L)$ from $NEG_n(L)$. Let q be an internal state of $\mathcal{A}$. We say that q is a *crossing state* if it is visited while $\mathcal{A}$ processes an input from $POS_n(L) \cup NEG_n(L)$ and the automaton passes through the cell containing the leftmost occurrence of $\#$. The *crossing size* of $\mathcal{A}$ is the number of its crossing states. We use $\|\mathcal{A}\|_c$ to denote the crossing size of $\mathcal{A}$.

Remark 40. The crossing states are the internal states that are used to carry information from one side (either ε or $r_1 \# \dots \# r_{\lfloor \log(n) \rfloor}$) to the other.

Definition 41. Let $L \subseteq \Sigma^*$. The nondeterministic crossing complexity of L is defined by:

$$2ncc(L, n) = \min \{ \|\mathcal{A}\|_c : \mathcal{A} \text{ is a 2NFA that separates } POS_n(L) \text{ from } NEG_n(L) \}.$$

We have:

Theorem 42. Let $L \subseteq \Sigma^*$. Let $\mathcal{A}_1$ be a nondeterministic pebble automaton separating $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$. Suppose the inequality

$$H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)) \leq f(n)$$

holds for infinitely many values of n . Then, there exists C such that the following inequality holds for infinitely many values of n :

$$2ncc(L, n) \leq 2^{f(n)} \cdot C \cdot \log(n).$$

Proof. Let $\mathcal{A}_1$ be a nondeterministic pebble automaton separating $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$ and such that the inequality

$$H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)) \leq f(n)$$

holds for infinitely many values of n . Let us fix such a n . Let $\mathcal{C}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)$ be the set of coding configurations that are visited by $\mathcal{A}_1$ when this automaton is processing an input from $\mathcal{H}_1^\Sigma(3 \cdot \lfloor \log(n) \rfloor \cdot (n+1))$. And let $\mathcal{CC}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)$ be the subset constituted by all the configurations visited by the automaton when it is asked to compute a *local bit* $w_i[1]$. We construct a 2NFA $\mathcal{M}_n$ such that:

1. $\mathcal{M}_n$ separates $POS_n(L)$ from $NEG_n(L)$.

2. The set of internal states of $\mathcal{M}_n$ is equal to:

$$\mathcal{C}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n) \times \{0, \dots, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1)\}^2 \times \{1, \dots, \log(n)\}^2.$$

3. The set of crossing states of $\mathcal{M}_n$ is included in

$$\mathcal{CC}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n) \times \{(\lfloor \log(n) \rfloor, 0)\} \times \{0, \dots, \log(n)\}^2.$$

Suppose $\mathcal{A}_1$ is provided with m -pebbles. Recall that an instantaneous configuration of $\mathcal{A}_1$ is a tuple $(h, (c_1, \dots, c_m, q))$ such that $h, c_i \in \{0, \dots, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1)\}$ and q is an internal state of $\mathcal{A}_1$.

Suppose:

- $\mathcal{A}_1$ is in the instantaneous configuration $(h, (c_1, \dots, c_m, q))$ scanning the local pair $w_i \#_1 r_i$.

Then:

- $\mathcal{M}_n$ is in the internal state $(c_1, \dots, c_m, q, (h, 0), (i, 0))$.

Moreover, we have:

- Let $(0, (0, \dots, 0, q_0))$ be the initial instantaneous configuration of $\mathcal{A}_1$.
The tuple

$$(0, \dots, 0, q_0, (0, 0), (0, 0))$$

is the initial state of $\mathcal{M}_n$.

- The set of accepting states of $\mathcal{M}_n$ is the set

$$\{(\vec{c}, q, (l, t) (i, j)) : (l, (\vec{c}, q)) \text{ is an accepting instantaneous configuration of } \mathcal{A}_1\}.$$

It only remains to define the transition function of $\mathcal{M}_n$. Let

$$s(w, r) = w_1 \#_1 r_1 \#_1 \dots \#_1 w_{\lfloor \log(n) \rfloor} \#_1 r_{\lfloor \log(n) \rfloor} \#_1^s$$

be an input of $\mathcal{A}_1$, and let

$$S(w, r) = w_1 [1] \dots w_{\lfloor \log(n) \rfloor} [1] \# r_1 \# \dots \# r_{\lfloor \log(n) \rfloor}$$

be the corresponding input of $\mathcal{M}_n$. The latter automaton simulates, on $S(w, r)$, the computation of $\mathcal{A}_1$ on $s(w, r)$. Let $(\vec{c}, q, (l, t) (i, j))$ be an internal state of $\mathcal{M}_n$. This automaton uses the coordinate l to track the location of its input head. The coordinate t is used as an auxiliary counter. On the other hand, $\mathcal{M}_n$ uses the last two components to track the factor being processed. Thus, suppose the input head of $\mathcal{A}_1$ is scanning the j -th letter of $\#_1 r_i$, and suppose that the current coding configuration of this pebble automaton is equal to $(\vec{c}, q)$. We assume that $\mathcal{M}_n$ is scanning the j -th letter of $\# r_i$ and that the internal state is equal to

$$(\vec{c}, q, (|r_1 \# \dots \# r_{i-1} \#| + j, 0), (i, 0)).$$

Notice that $\mathcal{M}_n$ can use the auxiliary counter represented by t to:

- check if the input head position matches with the position of a specific *virtual* pebble (i.e., matches $\vec{c}[i]$ for some $i \leq k$), and

- reset the contents of the first counter (i.e., the value of l) after this check.

Suppose $\mathcal{A}_1$ chooses to continue with the processing of r_i . The automaton $\mathcal{M}_n$ simulates this phase of the computation using the internal states in the set

$$\mathcal{C}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n) \times \{1, \dots, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1)\}^2 \times \{(i, 0)\}.$$

Notice that all those internal states of $\mathcal{M}_n$ are non-crossing states.

Now suppose that $(\vec{c}, q) \in \mathcal{CC}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)$ and that $\mathcal{A}_1$ chooses to scan the bit w_i [1]. Then, the automaton $\mathcal{M}_n$ has to move to the left in order to scan the i -th cell. Notice that, during this phase of the computation, $\mathcal{M}_n$ can set $t, j = 0$. Then, $\mathcal{M}_n$ can simulate this phase of the computation using the set

$$\{0, \dots, 3 \cdot \lfloor \log(n) \rfloor \cdot (n+1)\} \times \{0\} \times \{(\vec{c}, q)\} \times \{(i, 0)\}.$$

It can be verified that:

1. $\mathcal{M}_n$ separates $POS_n(L)$ from $NEG_n(L)$.
2. The set of crossing states of $\mathcal{M}_n$ is included in

$$\{(\lfloor \log(n) \rfloor + 1, 0)\} \times \mathcal{CC}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n) \times \{1, \dots, \lfloor \log(n) \rfloor\} \times \{0\}.$$

Let $C = |Q|$. The following inequalities hold for infinitely many values of n :

$$2ncc(L, n) \leq |\mathcal{CC}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)| \cdot C \cdot \log^2(n) \leq 2^{f(n)} \cdot C \cdot \log(n).$$

□

We obtain the following corollary:

Corollary 43. *Let $e, d > 0$, and suppose that the inequality $2ncc(L, n) \geq e \cdot n^d$ holds asymptotically. Then, for all $c < d$, all nondeterministic pebble automata $\mathcal{A}_1$ separating $L_1 \cap \mathcal{H}_1^\Sigma$ from $co-L_1 \cap \mathcal{H}_1^\Sigma$, and all $\gamma > 0$, the following inequality holds asymptotically:*

$$H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)) \geq c \cdot (1 - \gamma) \cdot \log(n).$$

Proof. Let $c < d$. Suppose that the inequality

$$H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)) \geq c \cdot (1 - \gamma) \cdot \log(n)$$

does not hold asymptotically for all $\gamma > 0$. Then, there exists $\gamma > 0$ such that the inequality

$$H(Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)) < d \cdot (1 - \gamma) \cdot \log(n)$$

holds for infinitely many values of n . Let n be one such value. The random variable $Y_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)$ is uniformly distributed over $\mathcal{CC}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)$. Then:

$$|\mathcal{CC}_{\mathcal{A}_1}(\mathcal{H}_1^\Sigma, n)| < 2^{\log(n^c)}.$$

We get that:

1. There exists C such that $2ncc(L, n) \leq n^c \cdot C \cdot \log(n)$ holds for infinitely many values of n .
2. The inequality $2ncc(L, n) \geq e \cdot n^d$ holds asymptotically.

We obtain a contradiction, the corollary follows. $\square$

Definition 44. Let $L \in \mathcal{CFL}$. We say that L is crossing-hard if there exists $c, d > 0$ such that, for all $n \geq 1$, the following inequality holds:

$$2ncc(L, n) > c \cdot n^d.$$

We have:

Corollary 45. *The following assertions hold:*

1. Let $L \in \mathcal{CFL}$ be crossing-hard. Then, the language L is hard-to-cross.
2. Let $L \in \mathcal{CFL}$. If L is crossing-hard the sequence $\{L_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy.
3. If there exists a context-free language that is crossing-hard the separation $\mathcal{NL} \neq \log \mathcal{CFL}$ holds.

8 Rectangles, digraph accessibility, and the language RA

Corollary 45 reduces the problem of proving the separation $\mathcal{NL} \neq \log \mathcal{CFL}$ to that of constructing a context-free language that is crossing-hard. We construct a context-free language, denoted RA, and show that it is crossing-hard. This language is closely related to the digraph accessibility problem.

Definition 46. Let G be a digraph. We say that G is a rectangle of depth k and width n if and only if:

1. $V(G) = \{1, \dots, n\} \times \{1, \dots, k+1\}$.
2. If $((i, j), (r, t)) \in E(G)$ the equality $t = j + 1$ holds.

Rectangles are *layered digraphs* with edges directed strictly between successive layers (levels). It is worth recalling that the restriction of digraph accessibility to the class of layered digraphs is $\mathcal{NL}$ -complete; see [2]. One can verify that the restriction of this latter problem to the class of rectangles is also $\mathcal{NL}$ -complete.

Definition 47. Let G be a rectangle of depth k . We say that G is crossable if there exists a path from the first level of G (i.e., the set $\{1, \dots, n\} \times \{1\}$) to the level $k + 1$.

Definition 48. Let G be a rectangle of depth k . We encode G as follows:

- Let $s < k + 1$ and let

$$l(s) = \{((i_1, s), (j_1, s + 1)), \dots, ((i_{m_s}, s), (j_{m_s}, s + 1))\}$$

be a list (which may include repetitions) of the edges that go from level s to level $s + 1$. We define:

$$r_{s, l(s)} = 1^{i_1} 0 1^{j_1} 0 \dots 0 1^{i_{m_s}} 0 1^{j_{m_s}}.$$

- We set $w_{G, l} = r_{1, l(1)} \$ \dots \$ r_{k, l(k)} \$$.

Let G be rectangle of depth k . Suppose $w_{G, l} = r_{1, l(1)} \$ \dots \$ r_{k, l(k)} \$$. Notice that R is the concatenation of k rectangles of depth 1, the rectangles encoded by the strings $r_{1, l(1)} \$, \dots, r_{k-1, l(k-1)} \$$, and $r_{k, l(k)} \$$.

Remark 49. Let n be a large integer. Let G be a rectangle of depth 1 and width $\lfloor n^{\frac{1}{4}} \rfloor$. This rectangle contains no more than $n^{\frac{1}{2}}$ edges. Furthermore, the code for each of these edges is no longer than $2n^{\frac{1}{4}} + 2$. Then there exists a code-word $w_{G,l}$ such that $|w_{G,l}| \leq n$, that is: we can think of $w \in \{0,1\}^n$ as the code of a rectangle of depth 1 and width $\lfloor n^{\frac{1}{4}} \rfloor$.

Definition 50. We use RA to denote the following context-free language

$$\{w_{G,l} \in \{0,1,\$ \}^* : G \text{ is crossable}\}.$$

We have:

Theorem 51. *RA is a context-free language.*

Proof. Let $w_1\$ \dots \$w_k\$$ be an instance of RA. This instance belongs to RA if and only if there exists factors $1^{i_1}01^{j_1}, \dots, 1^{i_k}01^{j_k}$ such that:

1. $1^{i_1}01^{j_1}$ is a factor of $w_1, \dots, 1^{i_{k-1}}01^{j_{k-1}}$ is a factor of w_{k-1} , and $1^{i_k}01^{j_k}$ is a factor of w_k .
2. $j_1 = i_2, \dots, j_{k-2} = i_{k-1}$, and $j_{k-1} = i_k$.

It is easy to build a nondeterministic pushdown automaton that guesses those factors and checks all those equalities. $\square$

We set $RA_\infty = L_{G_0}$, we have:

1. RA_∞ is an upper bound for $\{RA_k\}_{k \geq 0}$.
2. If RA is crossing-hard the sequence $\{RA_k\}_{k \geq 0}$ is high in the non-deterministic pebble hierarchy and the language RA_∞ belongs to $\mathcal{NL} - \log \mathcal{CFL}$.

8.1 Decorated versions of RA

From now, we study the crossing complexity of RA. We aim to show that RA is crossing-hard. Let us observe that this problem is a question about the *communication complexity* of 2NFAs.

Definition 52. Let $m \geq 1$. We define an injective function from $\{1, \dots, m\}$ to $\{0,1\}^m$. We define this injection as follows:

1. If m is even, we set

$$[i]_m = \begin{cases} (10)^i 0^{m-2i}, & \text{if } i \leq \frac{m}{2} \\ (11)^{i-\frac{m}{2}} (10)^*, & \text{otherwise} \end{cases}.$$

2. If m is odd, we set

$$[i]_m = \begin{cases} (10)^i 0^{m-2i}, & \text{if } i < \frac{m}{2} \\ (11)^{i-\lfloor \frac{m}{2} \rfloor} (10)^* 0, & \text{if } \frac{m}{2} < i < m \\ 1^m, & \text{if } i = m \end{cases}.$$

The next proposition is straightforward.

Proposition 53. *Suppose m is even. Then:*

1. $\lfloor \frac{m}{2} + 1 \rfloor_m = 11 (10)^{\frac{m}{2}-1}.$
2. $\lfloor \frac{m}{2} + 2 \rfloor_m = 1111 (10)^{\frac{m}{2}-2}.$
3. *For all $i \neq \frac{m}{2} + 1, \lfloor \frac{m}{2} + 2 \rfloor_m$, the string $[m]_i$ does not belong to $11 (10)^* \cup 1111 (10)^*$.*

Definition 54. We define:

- The set $POS_n^*(\text{RA})$ is equal to:

$$\left\{ \begin{array}{l} \varepsilon \# [1]_{\lfloor \log(n) \rfloor} \$ r_1 \# \cdots \# [\lfloor \log(n) \rfloor]_{\lfloor \log(n) \rfloor} \$ r_{\lfloor \log(n) \rfloor} : \\ \varepsilon \in \{0, 1\}^{\lfloor \log(n) \rfloor}, \quad r_1, \dots, r_{\lfloor \log(n) \rfloor} \in \{0, 1\}^{\leq n}; \\ \text{and } \odot_{\varepsilon[i]=1} r_i \$ \in \text{RA} \end{array} \right\}.$$

- The set $NEG_n^*(\text{RA})$ is equal to:

$$\left\{ \begin{array}{l} \varepsilon \# [1]_{\lfloor \log(n) \rfloor} \$ r_1 \# \cdots \# [\lfloor \log(n) \rfloor]_{\lfloor \log(n) \rfloor} \$ r_{\lfloor \log(n) \rfloor} : \\ \varepsilon \in \{0, 1\}^{\lfloor \log(n) \rfloor}, \quad r_1, \dots, r_{\lfloor \log(n) \rfloor} \in \{0, 1\}^{\leq n}; \\ \text{and } \odot_{\varepsilon[i]=1} r_i \$ \notin \text{RA} \end{array} \right\}.$$

Remark 55. It is important to observe that the elements of $POS_n^*(\text{RA}) \cup NEG_n^*(\text{RA})$ are decorated versions of the elements of $POS_n(\text{RA}) \cup NEG_n(\text{RA})$. We will show that those decorations do not increase the crossing complexity.

Let $\mathcal{M}$ be a 2NFA that separates $POS_n^*(RA)$ from $NEG_n^*(RA)$. The crossing states of $\mathcal{M}$ are the internal states visited when the automaton passes through the cell containing the leftmost occurrence of $\#$. We define $\|\mathcal{M}\|_c$ accordingly and we define $ncc(POS_n^*(RA), NEG_n^*(RA))$ as

$$\min \{ \|\mathcal{M}\|_c : \mathcal{M} \text{ separates } POS_n^*(RA) \text{ from } NEG_n^*(RA) \}.$$

We have:

Lemma 56. *For all $n \geq 1$, the following inequality holds:*

$$ncc(POS_n(RA), NEG_n(RA)) \geq ncc(POS_n^*(RA), NEG_n^*(RA)).$$

Proof. Let $\mathcal{M}$ be a 2NFA that separates $POS_n(RA)$ from $NEG_n(RA)$. This automaton can be used to separate $POS_n^*(RA)$ from $NEG_n^*(RA)$. It suffices if, on the input $\varepsilon \# [1]_{\lfloor \log(n) \rfloor} \$ r_1 \# \cdots \# [\log(n)]_{\lfloor \log(n) \rfloor} \$ r_{\lfloor \log(n) \rfloor}$, the automaton ignores the infixes

$$\# [1]_{\lfloor \log(n) \rfloor} \$, \dots, \# [\lfloor \log(n) - 1 \rfloor]_{\lfloor \log(n) \rfloor} \$, \text{ and } [\lfloor \log(n) \rfloor]_{\lfloor \log(n) \rfloor} \$.$$

The lemma follows from this. $\square$

We show, in Section 10, that there exist $c, d > 0$ such that, for all n large enough, the following inequality holds:

$$2ncc(POS_n^*(RA), NEG_n^*(RA)) \geq c \cdot n^d.$$

This implies that RA is crossing-hard and this also implies the separation $\mathcal{NL} \neq \log \mathcal{CFL}$.

9 On the crossing complexity of pattern matching

We are interested in the crossing complexity of digraph accessibility (the crossing complexity of the language RA). This leads us to study the crossing complexity of pattern matching:

1. We show that a promise problem related to pattern matching is crossing hard.
2. We show that the crossing complexity of this pattern matching problem is at most as high as the crossing complexity of RA.

Let $i \geq 1$. Recall that $\langle i \rangle$ denotes the binary code of $i - 1$.

Definition 57. Suppose $i \leq 2^{m+1}$. We define:

$$\langle i \rangle_m = 0^{m-|\langle i \rangle|} \langle i \rangle.$$

Definition 58. Let $n \geq 2$, we use PM_n to denote the following promise problem:

- Input: $\varepsilon \# 1^{n_1} \# \dots \# 1^{n_m}$, where $\varepsilon \in \{0, 1\}^{\lfloor \log(n) \rfloor}$; $m \leq 2^{4 \cdot \lfloor \log(n) \rfloor}$; and $n_1, \dots, n_m \leq 2^{\lfloor \log(n) \rfloor}$.
- Problem: decide whether there exists $i \leq m$ such that $\varepsilon = \langle n_i \rangle_{\lfloor \log(n) \rfloor}$.

We use PM_n^+ to denote the set of positive instances of PM_n . We use PM_n^- to denote the set of negative instances.

Remark 59. From now on, we use this notation with any promise problem under consideration, (i.e., L^+ and L^- denote the sets of positive and negative instances of the promise problem L).

We define $2ncc(PM_n^+, PM_n^-)$ in the usual way. We have:

Theorem 60. For all $n \geq 2$, the following inequality holds:

$$2ncc(PM_n^+, PM_n^-) \geq \left(\frac{n}{2}\right)^{\frac{1}{2}}.$$

Proof. Let $n \geq 2$ and let $w = \varepsilon \# 1^{n_1} \# \dots \# 1^{n_k}$ be an instance of PM_n . We use $S(w)$ to denote the set

$$\bigcup_{i \leq k} \{n_i\}.$$

which we call the *support* of w .

Let $\mathcal{M}_n$ be a 2NFA that solves the problem PM_n and let Q_c be its set of crossing states. Let $f_{w, \mathcal{M}_n} : Q_c \rightarrow 2^{Q_c}$ be the function defined by:

Let $p \in Q_c$, suppose the input head of $\mathcal{M}_n$ passes from left (ε) to right ($1^{n_1} \# \dots \# 1^{n_k}$) in the crossing state p , the set of crossing states that can be visited in the next pass (from right to left) is equal to $f_{w, \mathcal{M}_n}(p)$.

Suppose there exist two inputs $w = \varepsilon \# 1^{n_1} \# \dots \# 1^{n_k}$ and $u = \delta \# 1^{s_1} \# \dots \# 1^{s_t}$ such that:

- $S(w) \neq S(u)$, and
- $f_{w, \mathcal{M}_n} = f_{u, \mathcal{M}_n}$.

Let $j \notin S(w) \cap S(u)$ and let $\gamma = \langle j \rangle_{\lfloor \log(n) \rfloor}$. Set $w^* = \gamma \# 1^{n_1} \# \dots \# 1^{n_k}$ and $u^* = \gamma \# 1^{s_1} \# \dots \# 1^{s_t}$. Let us consider these two inputs. We have:

1. $w^* \in PM_n^+ \Leftrightarrow u^* \notin PM_n^+$
2. The automaton accepts both inputs or rejects both.

This is a contradiction. We conclude that there exists an injective function from $\mathcal{P}(\{1, \dots, 2^{\lfloor \log(n) \rfloor}\})$ into $(2^{Q_c})^{Q_c}$. This entails the inequality

$$\|\mathcal{M}_n\|_c \geq \left(\frac{n}{2}\right)^{\frac{1}{2}}.$$

The theorem follows. $\square$

In the next section, we show that a 2NFA separating $POS_n^*(RA)$ from $NEG_n^*(RA)$ requires as many states as a 2NFA solving the problem $PM_{n^{\frac{1}{4}}}$. This implies that RA is crossing-hard.

10 The language RA is crossing-hard

In this section, the last section of this work, we prove that RA is crossing-hard. We get as a corollary the separation $\mathcal{NL} \neq \log \mathcal{CFL}$.

10.1 Decorated versions of pattern matching

Definition 61. We use the following notation:

1. Let $x, y \in \{0, 1\}$, we use $x \oplus y$ to denote the XOR of x and y .
2. Let $\varepsilon = \varepsilon_1 \cdots \varepsilon_m \in \{0, 1\}^m$. We define:

$$\varepsilon^\oplus = \varepsilon_1 (1 \oplus \varepsilon_1) \cdots \varepsilon_m (1 \oplus \varepsilon_m).$$

Next proposition is straightforward.

Proposition 62. *Let ε and $\varepsilon^\oplus$ be as above. The pattern 111 does not occur in $\varepsilon^\oplus$.*

Consider the following *decorated version* of the problem $PM_{n^{\frac{1}{4}}}$.

Definition 63. Let $n \geq 1$. Let $X, Y \in \{0, 1, \#, \$\}^*$ be two strings such that X avoids the pattern $\#11(10)^*\$$ and Y avoids the pattern $\#1111(10)^*\$$. We use $LOC_n(X, Y)$ to denote the following promise problem.

- Input: $\varepsilon^{\oplus} 111X \# 11(10)^* \$ 1^{n_1} \mathbf{0} \dots \mathbf{0} 1^{n_m} \# 1111(10)^* \$ Y$,
where $\varepsilon \in \{0, 1\}^{\lfloor \frac{\log(n)}{4} \rfloor}$; $m \leq 2^{\lfloor \log(n) \rfloor}$; and $n_1, \dots, n_m \leq 2^{\lfloor \frac{\log(n)}{4} \rfloor}$.
- Problem: decide whether there exists an odd $i \leq m$ such that $\langle n_i \rangle_{\lfloor \frac{\log(n)}{2} \rfloor} = \varepsilon$.

Definition 64. Let $\mathcal{M}_n$ be a two-way automaton solving the problem $LOC_n(X, Y)$. Let q be an internal state of $\mathcal{M}_n$. We say that q is a crossing state if q is visited while the input head is scanning the leftmost occurrence of the pattern 111.

It is worth noting that instances of $LOC_n(X, Y)$ are nothing more than instances of $PM_{n^{\frac{1}{4}}}$ but decorated with an infix and a suffix that can be easily recognized. We have:

Lemma 65. *Let X and Y be as above. For all $n \geq 1$, the following inequality holds:*

$$2ncc(LOC_n^+(X, Y), LOC_n^-(X, Y)) \geq 2ncc\left(PM_{n^{\frac{1}{4}}}^+, PM_{n^{\frac{1}{4}}}^-\right).$$

Proof. Let $n \geq 1$ and let $\mathcal{M}_n$ be a nondeterministic pebble automaton that solves the promise problem $LOC_n(X, Y)$. Let $s(w) = \varepsilon \# 1^{n_1} \# \dots \# 1^{n_m}$ be an instance of $PM_{n^{\frac{1}{4}}}$. Suppose $\varepsilon \neq 0^*$. Let

$$S(w) = \varepsilon^{\oplus} 111X \# 11(10)^* \$ 1^{n_1} \mathbf{0} 1 \dots \mathbf{0} 1^{n_m} \mathbf{0} 1 \# 1111(10)^* \$ Y.$$

We construct a 2NFA $\mathcal{N}_n$ that, on input $s(w)$, simulates the computation of $\mathcal{M}_n$ on $S(w)$. We show that $\mathcal{N}_n$ separates $PM_{n^{\frac{1}{4}}}^+$ from $PM_{n^{\frac{1}{4}}}^-$ and that $\|\mathcal{M}_n\|_c = \|\mathcal{N}_n\|_c$.

Let Q be the set of internal of states of $\mathcal{M}_n$. The set of internal states of $\mathcal{N}_n$ is the set:

$$Q \times \{1, \dots, |X \# 11(10)^* \$| + |\# 1111(10)^* \$ Y| + 1\} \times \left\{1, \dots, \left\lfloor \frac{\log(n)}{4} \right\rfloor\right\}.$$

- Suppose $\mathcal{M}_n$ is in state q scanning the i -th position of $X \# 11(10)^* \$$. Then, $\mathcal{N}_n$ is in state $\left(q, i, \left\lfloor \frac{\log(n)}{4} \right\rfloor\right)$ with the input head placed on the cell $|\varepsilon \#| + 1$.

- Suppose $\mathcal{M}_n$ is in state q scanning the i -th position of the factor $1^{n_j}0$. Then, $\mathcal{N}_n$ is in state $\left(q, |X\#11(10)^*\$| + 1, \left\lfloor \frac{\log(n)}{4} \right\rfloor\right)$ with the input head placed on the i -th position of the factor $1^{n_j}0$.
- Suppose $\mathcal{M}_n$ is in state q scanning the i -th position of the factor $\#1111(10)^*\$Y$. Then, $\mathcal{N}_n$ is in state

$$\left(q, |X\#11(10)^*\$| + 1 + i, \left\lfloor \frac{\log(n)}{4} \right\rfloor\right)$$

with the input head placed on the last position of 1^{n_m} .

- Let $\delta = 0, 1$. Suppose $\mathcal{M}_n$ is in state q scanning the $(2i - \delta)$ -th position of the factor $\varepsilon^\oplus$. Then, $\mathcal{N}_n$ is in state $(q, 1, i)$ with the input head placed on the i -th position of ε .

The set of accepting states is the set

$$\{(q, i, j) : q \text{ is accepting}\}.$$

The transition function is defined accordingly. It is easy to check that $\mathcal{N}_n$ separates $PM_{n^{\frac{1}{4}}}^+$ from $PM_{n^{\frac{1}{4}}}^-$.

Let Q_c be the set of crossing states of $\mathcal{M}_n$. It is important to observe that the set of crossing states of $\mathcal{N}_n$ is equal to

$$Q_c \times \{1\} \times \left\{ \left\lfloor \frac{\log(n)}{4} \right\rfloor \right\}.$$

This implies the equality $\|\mathcal{M}_n\|_c = \|\mathcal{N}_n\|_c$. We obtain:

$$2ncc(LOC_n^+(X, Y), LOC_n^-(X, Y)) \geq 2ncc\left(PM_{n^{\frac{1}{4}}}^+, PM_{n^{\frac{1}{4}}}^-\right).$$

The lemma follows.

We obtain the following corollary: □

Corollary 66. *Let $X, Y \in \{0, 1, \#, \$\}^*$ be as above. For all $n \geq 2$, the following inequality holds:*

$$2ncc(LOC_n^+(X, Y), LOC_n^-(X, Y)) \geq \left(\frac{n}{2}\right)^{\frac{1}{8}}.$$

10.2 From pattern matching to rectangle traversing

Let $n \geq 2$. We show that there exist two strings $X_n, Y_n \in \{0, 1, \#, \$\}^*$ such that:

1. X_n avoids the pattern $\#11(10)^*\$$ and Y_n avoids the pattern $\#1111(10)^*\$$.
2. If $\mathcal{M}$ is a 2NFA that separates $POS_n^*(\text{RA})$ from $NEG_n^*(\text{RA})$, then this same automaton solves the *promise problem* $LOC_n(X_n, Y_n)$.

We construct X_n and Y_n in such a way that every positive instance of $LOC_n(X_n, Y_n)$, say

$$Z = \varepsilon^{(2)}111X_n\#11(10)^*\$1^{n_1}0 \dots 01^{n_m}\#1111(10)^*\$Y_n,$$

is an element of $POS_n^*(\text{RA})$ and every negative instance is an element of $NEG_n^*(\text{RA})$

1. If we view Z as an instance of $LOC_n(X_n, Y_n)$, the problem we are considering is the problem of deciding whether we can find the pattern ε within the set $\{n_1, \dots, n_m\}$. Here, ε acts as a pattern.
2. If we view Z as an element of $POS_n^*(\text{RA}) \cup NEG_n^*(\text{RA})$, the problem we are considering is the problem of deciding whether the rectangle encoded by Z can be crossed from left to right. Here, ε acts as part of a mask.

Then, we have to build a gadget that converts masks into patterns. Given $n \geq 32$, we construct a gadget B_n that fulfills this function.

Definition 67. Let $n \geq 32$. The construction of B_n proceeds as follows:

1. Given $1 \leq i \leq \left\lfloor \frac{\log(n)}{4} \right\rfloor$, we choose a word $r_{2i-1} \in \{0, 1\}^{\leq n}$ encoding the rectangle $G_{i,1}$ that is defined by:

$$\begin{aligned} \bullet V(G_{i,1}) &= \left\{ 1, \dots, 2^{\left\lfloor \frac{\log(n)}{4} \right\rfloor} \right\} \times \{1, 2\}. \\ \bullet E(G_{i,1}) &= \left\{ ((s, 1), (s, 2)) : \text{the } i\text{-th bit of } \langle s \rangle_{\left\lfloor \frac{\log(n)}{4} \right\rfloor} \text{ is equal to } 1 \right\}. \end{aligned}$$

2. Given $1 \leq i \leq \left\lfloor \frac{\log(n)}{4} \right\rfloor$, we choose a word $r_{2i} \in \{0, 1\}^{\leq n}$ that encodes the even version of $G_{i,1}$, namely the rectangle $G_{i,0}$ defined by the following dual condition:

$$E(G_{i,0}) = \left\{ ((s, 1), (s, 2)) : \text{the } i\text{-th bit of } \langle s \rangle_{\left\lfloor \frac{\log(n)}{4} \right\rfloor} \text{ is equal to } 0 \right\}.$$

3. We set:

$$B_n = r_1 \$ \cdots \$ r_{2^{\lfloor \frac{\log(n)}{4} \rfloor}} \$.$$

Definition 68. Let $\varepsilon \in \{0, 1\}^{\lfloor \frac{\log(n)}{4} \rfloor}$. We use i_ε to denote the integer in $\left\{1, \dots, 2^{\lfloor \frac{\log(n)}{4} \rfloor}\right\}$ that satisfies the condition $\langle i_\varepsilon \rangle_{\lfloor \frac{\log(n)}{4} \rfloor} = \varepsilon$.

Lemma 69. Let $\varepsilon \in \{0, 1\}^{\lfloor \frac{\log(n)}{4} \rfloor}$ and let $B_n(\varepsilon)$ be the rectangle encoded by the string

$$\bigodot_{i \leq 2^{\lfloor \frac{\log(n)}{4} \rfloor} : \varepsilon^\oplus[i]=1} r_i \$.$$

The only node in the last level of $B_n(\varepsilon)$ that can be reached from the first level is the node $\left(i_\varepsilon, \left\lfloor \frac{\log(n)}{4} \right\rfloor + 1\right)$.

Proof. Let $\varepsilon \in \{0, 1\}^{\lfloor \frac{\log(n)}{4} \rfloor}$. Notice that:

1. The rectangle $B_n(\varepsilon)$ is the rectangle of depth $\lfloor \log(n)/4 \rfloor$ encoded by the string

$$\bigodot_{i \leq \lfloor \log(n)/4 \rfloor} G_{i, \varepsilon[i]} \$.$$

2. Suppose that there exists a path from the first layer of $B_n(\varepsilon)$ to the node $\left(s, \left\lfloor \frac{\log(n)}{4} \right\rfloor + 1\right)$. This path is equal to

$$(s, 1), (s, 2), \dots, \left(s, \left\lfloor \frac{\log(n)}{4} \right\rfloor + 1\right).$$

3. For all $i \leq \left\lfloor \frac{\log(n)}{4} \right\rfloor$ and all $s \in \left\{1, \dots, 2^{\lfloor \frac{\log(n)}{4} \rfloor}\right\}$, the edge $((s, i), (s, i+1))$ occurs in $B_n(\varepsilon)$ if and only if

$$\varepsilon[i] = \langle s \rangle_{\lfloor \frac{\log(n)}{4} \rfloor}[i].$$

The lemma follows from these three facts. □

10.3 Main theorem

We have:

Theorem 70. *For all $n \geq 2^{12}$, the following inequality holds:*

$$2ncc(RA, n) \geq \left(\frac{n}{2}\right)^{\frac{1}{8}}.$$

Proof. We show that, for all $n \geq 2^{12}$,

$$2ncc(POS_n^*(RA), NEG_n^*(RA)) \geq \left(\frac{n}{2}\right)^{\frac{1}{8}}.$$

Remark 71. If $n \geq 2^{12}$, the inequality $\left\lfloor \frac{\log(n)}{4} \right\rfloor - 3 \geq 0$ holds.

Let us suppose that $\lfloor \log(n) \rfloor$ is divisible by 4.

1. Let $\varepsilon = \varepsilon_1 \cdots \varepsilon_{\lfloor \frac{\log(n)}{4} \rfloor} \in \{0, 1\}^{\lfloor \frac{\log(n)}{4} \rfloor}$. Let $r_{\lfloor \frac{\log(n)}{2} \rfloor + 2}, \dots, r_{\lfloor \log(n) \rfloor} \in \{0, 1\}^{\leq n}$ be words such that, for all $\frac{\lfloor \log(n) \rfloor}{2} + 1 \leq 2 \leq \lfloor \log(n) \rfloor$, the string r_i encodes a *complete* rectangle of depth 2 and width $2^{\frac{\lfloor \log(n) \rfloor}{4}}$ (i.e., $E(G_i) = \left\{1, \dots, 2^{\frac{\lfloor \log(n) \rfloor}{4}}\right\}^2$).
2. Let $r_1, \dots, r_{\lfloor \frac{\log(n)}{2} \rfloor}$ be like in the construction of the gadget B_n , see Definition 67.
3. Define:
 - $X_n = 1^{\frac{\lfloor \log(n) \rfloor}{4} - 3} \# [1]_{\lfloor \log(n) \rfloor} \$r_1 \# \cdots \# \left[\frac{\lfloor \log(n) \rfloor}{2} \right]_{\lfloor \log(n) \rfloor} \$r_{\lfloor \frac{\log(n) \rfloor}{2}},$
and
 - $Y_{n+1} = \left[\frac{\lfloor \log(n) \rfloor}{2} + 2 \right]_{\lfloor \log(n) \rfloor} \$r_{\lfloor \frac{\log(n) \rfloor}{2} + 2} \# \cdots \# [\lfloor \log(n) \rfloor]_{\lfloor \log(n) \rfloor} \$r_{\lfloor \log(n) \rfloor}.$

Notice that:

1. $\left[\frac{\lfloor \log(n) \rfloor}{2} + 1 \right]_{\lfloor \log(n) \rfloor} = 11 (10)^{\frac{\lfloor \log(n) \rfloor}{2} - 1}.$
2. $\left[\frac{\lfloor \log(n) \rfloor}{2} + 2 \right]_{\lfloor \log(n) \rfloor} = 1111 (10)^{\frac{\lfloor \log(n) \rfloor}{2} - 2}.$

3. Let $n_1, \dots, n_m \leq 2^{\frac{\lfloor \log(n) \rfloor}{4}}$ and $m \leq 2^{\log(n)}$. Let

$$Z = \varepsilon^{\oplus} 111X_n \# 11(10)^* \$ \mathbf{1}^{n_1} \mathbf{0} \dots \mathbf{01}^{n_m} \$ \# 1111(10)^* \$ Y_n.$$

It follows, from Proposition 53, that $\#11(10)^* \$$ does not occur in X_n and that $\#1111(10)^* \$$ does not occur in Y_n . Thus, we have that the promise problem $LOC_n(X_n, Y_n)$ is well defined and that Z is an instance of $LOC_n(X_n, Y_n)$.

4. Let

$$Z = \varepsilon^{\oplus} 111X_n \# 11(10)^* \$ \mathbf{1}^{n_1} \mathbf{0} \dots \mathbf{01}^{n_m} \# 1111(10)^* \$ Y_n$$

be an instance of $LOC_n(X_n, Y_n)$. Suppose $\varepsilon \neq 0^*$. If m is even, we set $Z^* = Z$. If m is odd, we set:

$$Z^* = \varepsilon^{\oplus} 111X_n \# 11(10)^* \$ \mathbf{1}^{n_1} \mathbf{0} \dots \mathbf{01}^{n_m} \mathbf{01} \# 1111(10)^* \$ Y_n.$$

We notice that Z is a positive instance of $LOC_n(X_n, Y_n)$ if and only if Z^* is a positive instance of this promise problem.

5. Let

$$Z = \varepsilon^{\oplus} 111X_n \# 11(10)^* \$ \mathbf{1}^{n_1} \mathbf{0} \dots \mathbf{01}^{n_m} \# 1111(10)^* \$ Y_n$$

be an instance of $LOC_n(X_n, Y_n)$. We can assume that m is even. We assume this. Then, Z is an element of $POS_n^*(\text{RA}) \cup NEG_n^*(\text{RA})$.

6. Let Z be like in the previous item (i.e., m is even). It follows from Lemma 69 that Z belongs to $POS_n^*(\text{RA})$ if and only if there exists an odd $i \leq m$ such that $\varepsilon = \langle n_i \rangle_{\frac{\lfloor \log(n) \rfloor}{4}}$.
7. Let Z be like in the previous two items. We have that Z is a positive instance of $LOC_n(X_n, Y_n)$ if and only if Z belongs to $POS_n^*(\text{RA})$.

Then, if $\mathcal{M}_n$ is a 2NFA that separates $POS_n^*(\text{RA})$ from $NEG_n^*(\text{RA})$ this automaton solves the promise problem $LOC_n(X_n, Y_n)$. The following inequalities hold:

$$\|\mathcal{M}_n\|_c \geq \left(2^{\frac{\lfloor \log(n) \rfloor}{4}}\right)^{\frac{1}{2}} \geq \left(\frac{n}{2}\right)^{\frac{1}{8}}.$$

The theorem is proved. □

Corollary 72. *The following assertions hold:*

1. *The language RA is crossing-hard.*
2. *The language RA is hard-to-cross.*
3. *Let $\Sigma = \{0, 1, \$\}$. For all $d < \frac{1}{8}$, all $\gamma > 0$, and all nondeterministic pebble automaton accepting the language RA_k , the following inequality holds asymptotically:*

$$H(X_{\mathcal{A}_k}(\mathcal{H}_k^\Sigma, n)) \geq k \cdot d \cdot (1 - \gamma) \cdot \log(n).$$

4. *The sequence $\{RA_k\}_{k \geq 0}$ is high in the nondeterministic pebble hierarchy and the language $\bar{RA}_\infty = L_{G_0}$ belongs to $\mathcal{NL} - \log \mathcal{CFL}$.*
5. *The separations $\mathcal{NL} \neq \log \mathcal{CFL}$ and $\mathcal{NL} \neq \mathcal{P}$ hold.*

Acknowledgement. The author thanks Centro de Investigaciones Antonio Sánchez de Cozar, vereda los Egidos, San Gil, Colombia. Most of this work was written while the author was visiting this charming research center in Santander mountains. Dedicated to Octavio, Marina, Carolina, the children, and the dogs.

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